

Vidya Sahayak — Multilingual Voice-First AI Tutor for Tier-2/3 India

Bharat Academix CodeQuest 2026 · Education Technology Track

Team Members

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Thank you to the Bharat Academix organizers for hosting this hackathon.

We are deeply motivated to turn Vidya Sahayak
into a real solution for students across Bharat.

Agenda

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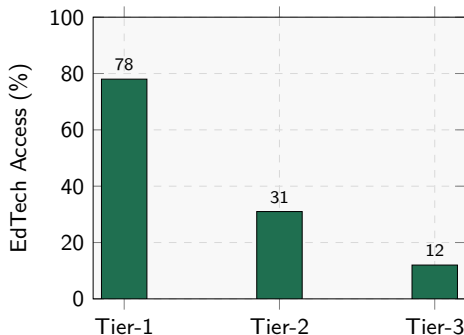
The Problem: 300M Students, Zero Voice in Their Language

The Doubt Wall

A student in Patna, Madurai, or Siliguri encounters a complex concept. The only resources available are:

- ✗ English-only explanations
- ✗ Text-only, no voice interaction
- ✗ Generic analogies (Wall Street, not *kirana* shops)
- ✗ Designed for metro-school context
- ✗ Assumes high-bandwidth connectivity

Existing platforms (Byju's, Khan Academy, ChatGPT) treat regional language as a translation afterthought — not a first-class experience.



Source: estimated access gap, Tier-wise.

Problem: Root Causes

Language Barrier

- 90%+ online STEM content is in English
- Hindi, Tamil, Bengali support is surface-level
- Scripts rendered as images, not native text

Interaction Barrier

- Text typing is slow & unfamiliar on budget Androids
- No voice-first tutor exists for Indian regional languages
- Students naturally ask doubts *aloud*

Context Barrier

- Foreign analogies don't land in Indian classrooms
- No grade-level calibration (Class 6 $\neq$ Class 10)
- No way to know if the AI is confident or guessing

Infrastructure Barrier

- Patchy 2G/3G connectivity in Tier-2/3 towns
- Heavy apps time out or drain data
- Audio auto-play is a luxury, not a default

Introducing Vidya Sahayak ()

A voice-first, multilingual AI tutor

that speaks to Indian students in *their* language,
using *their* everyday analogies,
on *their* bandwidth.



4 Languages

Hindi, Tamil, Bengali, English
(6 more coming)



Voice-First

Speak your doubt,
get a spoken answer



AI-Powered

Gemini 2.5 Flash
with honest confidence scores

6 Core Features at a Glance

#	Feature	What judges see	Status
1	Voice doubt-solving	Speak Hindi/Tamil/Bengali, hear answer back	LIVE
2	Tuition-Teacher Mode	Cricket/kirana-shop analogies on toggle	LIVE
3	Doubt-to-Diagram	SVG diagram renders for science/math doubts	HYBRID
4	Confidence Calibration	High/Medium/Low badge, verified vs. fact-keys	LIVE
5	Weak-Topic → Practice	5-MCQ quiz generated after 3 related doubts	LIVE
6	Low-Bandwidth Mode	Short answers, manual audio, same quality	REAL

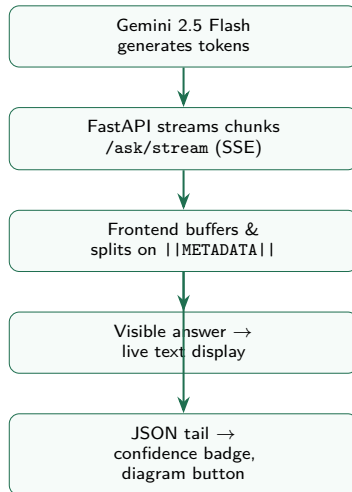
Innovation 1 — Streaming Answer Pipeline (SSE)

What it is

The answer streams token-by-token to the frontend via Server-Sent Events (SSE). The user sees text appearing live *while* Gemini is still generating, instead of a blank loading screen.

How it works

- FastAPI `/ask/stream` yields chunks
- Custom `|||METADATA|||` delimiter separates visible answer from backend JSON metadata
- Frontend splits stream on delimiter to extract confidence, `topic_tag`, `diagram_eligible` without an extra round-trip



Innovation 2 — Confidence Calibration with Fact-Key Verification

The Problem with Self-Reported AI Confidence

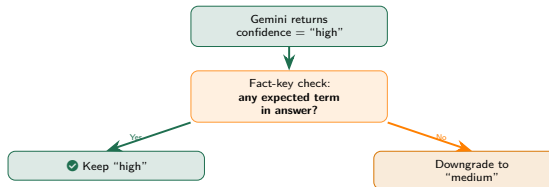
LLMs confidently state wrong answers. Simply asking Gemini “how confident are you?” is insufficient.

Our Two-Layer Solution

Layer 1: Gemini self-reports high / medium / low

Layer 2: Backend `verify_confidence()` cross-checks the answer text against a curated **fact-key dictionary** per topic per language:

- Photosynthesis in Hindi → must contain , , etc.
- If none match and model said “high” → **automatically downgraded to “medium”**
- *Only ever downgrades, never upgrades* — absence of a downgrade is not proof of correctness



Covers 6 core topics × 4 languages = 24 fact sets

Innovation 3 — “Explain Like My Tuition Teacher” Mode

The Insight

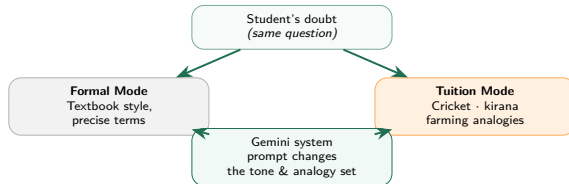
The best tutors in India don't explain like textbooks — they use cricket matches, *kirana* shop transactions, train journeys, and cooking analogies. No AI tutor has operationalized this.

Formal Mode (default)

Photosynthesis is the process by which plants synthesize carbohydrates from carbon dioxide and water using solar energy in the presence of chlorophyll. . .

Tuition Mode

“ ” , CO₂ ()



One-tap re-explain button re-submits the same question with the opposite mode prompt.

Innovation 4 — Doubt-to-Diagram (Hybrid Rendering)

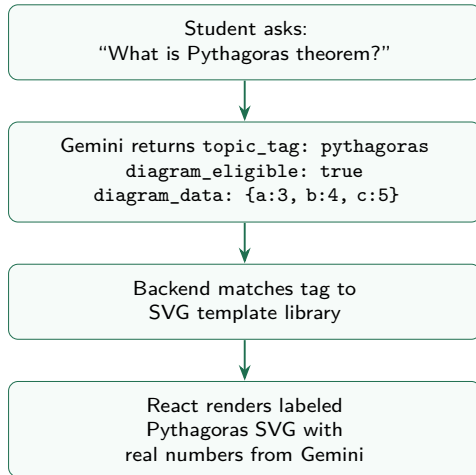
Design Rationale

Full generative SVG from an LLM is unreliable for diagrams. We **pre-build 6 accurate SVG templates** and let Gemini populate the data values dynamically.

Supported diagram topics:

- Photosynthesis (chloroplast process flow)
- Water Cycle (evaporation → precipitation)
- Pythagoras Theorem (with live a, b, c values)
- Fractions (pie-slice with numerator/denominator)
- Simple Electrical Circuit
- Human Digestive System

Gemini returns `diagram_eligible: true` + optional `diagram_data` (e.g., `{side_a:3, side_b:4, hypotenuse:5}`) for live injection into the template.



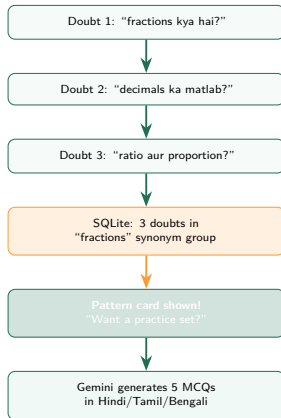
Innovation 5 — Weak-Topic Tracker + Auto Practice Set

The Retention Problem

Answering a doubt once doesn't mean the student understands. Repeated doubts on related topics signal a genuine knowledge gap.

How it works:

- 1 Every question is tagged and logged to SQLite with `session_id`, `topic_tag`, `language`, `timestamp`
- 2 A **synonym group map** clusters related tags (fractions / decimals / ratios / percentages = same group)
- 3 After **3 doubts** in the same synonym group → "We noticed a pattern" card appears
- 4 One tap → `/practice` endpoint calls Gemini to generate **5 MCQs** in the student's language
- 5 Instant right/wrong feedback, trophy score at the end



Innovation 6 — Hinglish / Romanized Input Detection

Real-World Usage Pattern

Students on budget Android keyboards often type their Hindi/Tamil/Bengali question using **English (Roman) script** — “photosynthesis kya hai”, “prakaash sanshllleshan explain karo”. Standard NLP models fail here.

Our detection heuristic:

- 1 If selected language $\neq$ English
- 2 **AND** the question text is pure ASCII (no native Unicode script characters)
- 3 $\Rightarrow$ Flagged as Romanized input
- 4 A special prefix is prepended to the Gemini system prompt:
“The student typed in Romanized Hindi. Understand it as Hindi and answer entirely in Devanagari script.”

Example

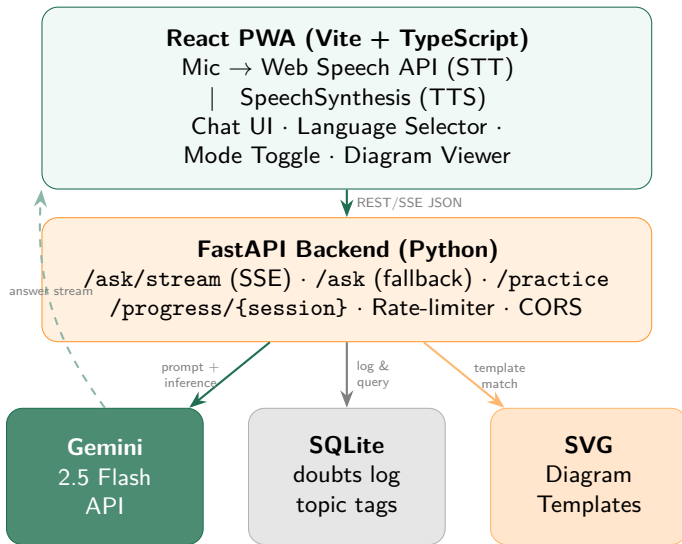
Input (Roman, language=hi):
prakash sanshllleshan kya hai

Output (Devanagari): ...

Why This Matters

No manual transliteration API needed. Zero latency overhead. Works entirely via a prompt injection before the LLM call. Pure ASCII check is **O(n)** and branch-free.

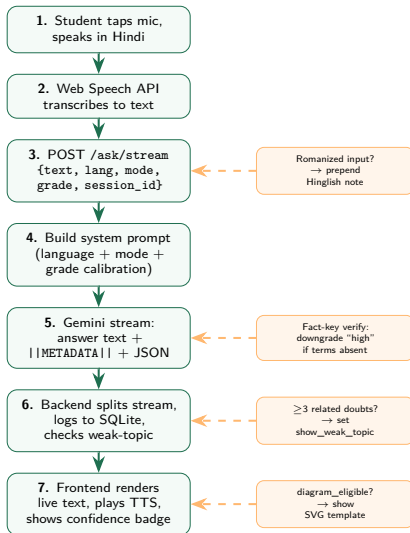
High-Level Architecture



Tech Stack at a Glance

Layer	Choice	Rationale
Frontend	React 18 + Vite + TypeScript	Fast HMR, strict typing, PWA-ready
STT / TTS	Browser Web Speech API	Free, native, no external service, works offline-ish
Backend	FastAPI (Python 3.11)	Async first-class, Pydantic validation, minimal setup
LLM	Google Gemini 2.5 Flash	Best cost-performance for flash inference; streaming SDK
Database	SQLite	Zero-config, sufficient for MVP/demo, single-file deploy
Diagrams	Pre-built React SVG components	Reliable rendering; dynamic data injection from Gemini
State	React Context + localStorage	Persistent session, language, mode, grade, bandwidth
Hosting	Vercel (FE) + Render (BE)	Free-tier, instant deploy, accessible to judges live

End-to-End Data Flow: Voice Doubt → Answer



Database & Session Model

SQLite Schema

Table: doubts

Column	Description
id	Auto-increment PK
session_id	UUID (per device/browser)
text	Raw question text
topic_tag	Gemini-assigned tag
language	hi / ta / bn / en
timestamp	UTC ISO-8601

Session ID Strategy

- Generated as UUIDv4 in the browser on first load
- Persisted to localStorage
- Sent with every API request
- Enables per-student progress tracking *without auth*
- Demo uses single session; production would add real auth

Synonym Group Map

10 synonym groups cover 40+ topic tags. E.g. {fractions, decimals, ratios, percentages} all count toward the *same* weak-topic threshold.

UI/UX Design Principles

Mobile-First, Tap-Target-First

- Giant mic button = primary action, impossible to miss
- Language selector uses native scripts prominently (, ,) — not just English names
- Grade selector (Class 6–10) calibrates Gemini output complexity live
- Bottom nav with 3 tabs: Ask / Progress / Settings

Deliberately Not Silicon Valley Blue

- Deep green #1F6F50 + saffron accents
- Warm, approachable — feels Indian-EdTech, not SaaS
- Devanagari/Tamil/Bengali rendered natively in answer body
- Confidence badge: green pill / amber pill / red pill with teacher warning on “low”

Accessibility & Inclusion

- `aria-label` on all interactive controls
- `Role dialog + aria-modal` on overlays
- Keyboard navigation: Enter submits question
- `role=alert` on error messages for screen readers
- Disabled states clearly communicated via attribute

Settings Persistence

Language, mode, grade, and bandwidth setting are saved to `localStorage` and restored on next visit — zero friction return experience.

Backend Robustness Features

Resilience

- **3-attempt retry with exponential backoff** on every Gemini call (`ask_gemini`, `ask_gemini_stream`, `ask_gemini_practice`)
- **Per-request timeout** (default 10s) via `asyncio.wait_for` — no hanging requests
- Graceful JSON parse fallback: if Gemini returns malformed JSON, the answer text is still served with confidence: "low" rather than crashing

Security & Rate Limiting

- **IP-based sliding window rate limiter** (120 req/60s per IP, configurable via env)
- **Request size cap** (64 KB, configurable) — prevents prompt-stuffing attacks
- **CORS restricted** to `localhost:5173` by default; production origin set via env
- **Parameterized SQLite queries** throughout — no SQL injection surface
- `GEMINI_API_KEY` never leaves the backend

Prompt Engineering — Grade-Calibrated, Multi-Mode

System prompt layers (stacked):

- ① **Language rule** — strict native-script-only output
- ② **Style rule** — formal vs. tuition-teacher analogies
- ③ **Length rule** — 3–6 sentences, or 40-word cap (low-bandwidth)
- ④ **Grade calibration** — Class 6–7: simple everyday language; Class 10: full terminology, exam-level depth
- ⑤ **Confidence rule** — self-report high/medium/low
- ⑥ **Topic classification** — snake-case tag
- ⑦ **Diagram rule** — flag eligible topics
- ⑧ **Diagram data rule** — inject real numbers for Fractions and Pythagoras templates
- ⑨ **Response format** — exact delimiter format
|||METADATA|||

Why this design is robust

- Each instruction is on its own numbered/labelled line — easy for the LLM to parse and obey
- Format instruction includes **concrete examples** of correct output for every case (with and without diagram_data)
- Delimiter is unique enough (|||METADATA|||) that it won't accidentally appear in an answer
- Temperature: **0.4** for answers (focused), **0.6** for practice questions (diverse options)
- Token budgets: 1024 for answers, 2048 for practice

Rubric Alignment — How We Score on Every Criterion

Criterion	Weight	Our Differentiator
Innovation & Creativity	25%	Confidence calibration + fact-key verification; tuition-teacher analogy mode; Romanized input detection — none of these are in a standard AI chatbot
Technical Implementation	25%	Real streaming SSE pipeline; SQLite-backed weak-topic tracker; React SVG diagram templates with live data injection from Gemini; production-grade retry & rate-limiting
Problem-Solving Approach	20%	Every feature maps directly to a named barrier (language, interaction, context, infrastructure) — not a generic AI wrapper
UX & Design	15%	Voice-first; native script rendered prominently; warm Indian-EdTech palette; accessible ARIA labels; low-bandwidth mode; grade selector
Scalability & Impact	15%	Language-pack architecture: adding a language is a config change; directly addresses 300M+ underserved students; open to Gujarati, Marathi, Telugu, Kannada. . .

Competitive Differentiation

Feature	Byju's	Khan Academy	ChatGPT	Vidya Sahayak
Regional language (first-class)	Partial	No	No	✓
Voice doubt input	No	No	No	✓
Spoken answer in regional lang.	No	No	No	✓
Tuition-teacher analogy mode	No	No	No	✓
Confidence calibration (verified)	No	No	No	✓
Weak-topic detection + MCQ	Partial	Partial	No	✓
Low-bandwidth mode	No	No	No	✓
Romanized (Hinglish) input support	No	No	Partial	✓
Grade-level answer calibration	Partial	Partial	No	✓

Responsible AI by Design

Honest About Uncertainty

- Every answer carries a visible confidence badge
- “Low” confidence triggers a teacher-verification nudge
- Two-layer confidence check (self-report + fact-key verification) reduces silent hallucinations
- No pretense of being a replacement for a teacher

Transparent Hardcode Map

We document exactly what is AI-generated vs. templated: diagram SVGs are pre-built; weak-topic threshold is rule-based; session auth is demo-only.

“We scoped this consciously” is more credible than pretending everything is fully built.

Privacy & Security

- No PII collected — session ID is an anonymous UUID
- API key never exposed to frontend
- Rate limiting prevents abuse
- Request size cap prevents prompt injection
- CORS restricted by default
- All DB queries parameterized

Inclusion by Default

Low-bandwidth mode isn't a premium feature toggle — it's a first-class setting on the Settings screen, treated as equally important as language selection.

Scalability by Design

Language-Pack Architecture

Adding a new Indian language requires:

- ➊ Add language code to `LANGUAGE_NAMES` dict
- ➋ Add fact-keys to `VERIFIED_FACT_KEYS`
- ➌ Add synonym group entries
- ➍ Update browser STT `lang` attribute
- ➎ Add UI pill in language selector

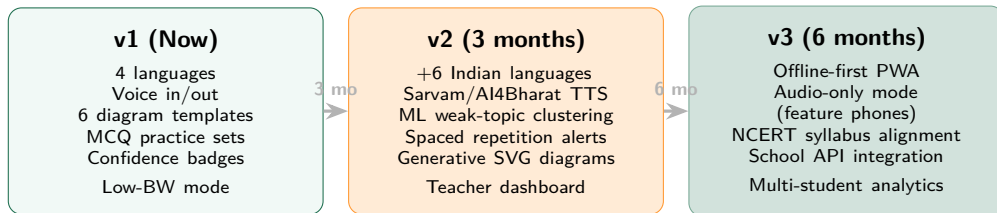
No new backend endpoints, no new models.

Gujarati, Marathi, Telugu, Kannada, Punjabi, Odia are already listed in the Settings UI as “Coming Soon”.

Infrastructure Scaling

- **SQLite** → **PostgreSQL** for multi-instance production (schema is already migration-ready)
- **Gemini API** scales with quota increases, no infra change needed
- **FastAPI** is async-native — handles concurrent streaming sessions efficiently
- **Frontend** is a static PWA — CDN-deployable, near-zero latency globally
- Rate limiter is currently in-memory; swap to Redis for distributed deployments

Product Roadmap



Demo Script — 5 Minutes, 7 Moments

- 1 **Open the app** — show the Indian-palette UI, language pills with Devanagari/Tamil/Bengali scripts, the giant mic button
- 2 **Voice doubt in Hindi** — tap mic, speak “*prakash sanshlleshan kya hai*”, get a spoken answer back in Hindi. *This is your hook moment.*
- 3 **Flip the mode toggle live** — tap “Tuition Style”, tap re-explain. Show the kirana-shop analogy version side by side. Judges *visibly react* to the tone difference.
- 4 **Diagram doubt** — ask “*Pythagoras theorem explain karo*”, show the SVG diagram render with real triangle values. *Visual proof of technical depth.*
- 5 **Confidence badge** — ask a harder/ambiguous question, point out the amber or red badge. Frame as a *responsible AI design choice*, not a limitation.
- 6 **Weak-Topic card** — ask 3 fraction-related doubts, show the pattern card, tap “Generate Practice Set”, complete one MCQ. *“We thought about retention, not just answering.”*
- 7 **Low-bandwidth toggle** — flip in Settings, show shorter answer, manual audio tap. Ties back to the Tier-2/3 framing.

Anticipated Judge Questions — Honest Answers Ready

Question	Our answer
Why only 6 diagram templates?	Pre-built for MVP speed. Full generative SVG is the v2 milestone. Templates cover 80% of NCERT Class 6–10 diagram doubts.
Is the weak-topic threshold trained?	Rule-based for now (3 doubts = threshold). An ML topic-clustering model with spaced repetition is the v2 upgrade.
How reliable is the confidence score?	Two layers: Gemini self-reports, then we cross-check against a fact-key dictionary. Only ever downgrades — never falsely promotes.
What about Marathi/Telugu?	Architecture is language-pack based. Adding a language is a config change. 6 languages are listed as Coming Soon in the Settings UI.
What replaces Gemini if the API breaks?	Retry with exponential backoff is built in. A model-agnostic LLM client layer makes swapping to GPT-4o or Llama a one-line change.

Vidya Sahayak

A voice-first, multilingual AI tutor
that speaks to India's 300 million students
in their language, using their analogies,
on their bandwidth.



Hindi



Tamil



Bengali

Built for Bharat Academix CodeQuest 2026

Stack: React · FastAPI · Gemini 2.5 Flash · SQLite · Web Speech API



github.com/your-team/vidya-sahayak